Administrator (extends User)

Responsibilities	Collaborators
Browse all campsites	Campsite
Remove a campsite	
Browse all lotteries	Lottery
Remove a lottery	
Browse all user profiles	User
Remove a User profile	
Browse all uploaded images	Image
Remove an image	
Remove a composite owner’s privileges	AdminController

User

Responsibilities	Collaborators
Know name, email, phone #, device ID (01.09.01)	
Know its profile picture	Image
Can Delete profile	ProfilesStorage

ImageStorage

Responsibilities	Collaborators
Return all stored images	Image
Delete image from storage	Firebase storage

Image

Responsibilities	Collaborators
Upload images to cloud storage	Image
Delete images	Firebase Storage

Return image URLs	

CampsitesStorage

Responsibilities	Collaborators
Returns stored campsites	Campsite
Deletes a campsite from storage	
Adds a campsite	

Campsite

Responsibilities	Collaborators
Know its availability	Lottery
Know name, price per night, max occupancy, rating	Campsite Owner
Know rules, cancellation policy, amenities	
Know its Owner	
Know its location	Geolocation
Know its promotional image	image
Know its availability	Lottery
Know its lottery winners	
Know its published/draft state	

ProfilesStorage

Responsibilities	Collaborators
Returns stored profiles	Profile
Deletes profile	
Adds a profile	

Profile

Responsibilities	Collaborators

MessageService

Responsibilities	Collaborators
	Message
	User

Message

Responsibilities	Collaborators

QRCode

Responsibilities	Collaborators
Shows campsite details	Campsite
Scan a QR code with the camera	
Open the matching campsite listing	

AvailabilitySchedule

Responsibilities	Collaborators
Store available date ranges for a campsite	Campsite
Check whether a date range is available	Date range
Mark dates as booked when a reservation is confirmed	Reservation
Free dates when a reservation is cancelled	
Provide data for the calendar view	

DateRange

Responsibilities	Collaborators

Know start and end dates	
Compute number of nights	
Check overlap with another date range	

WaitingList

Responsibilities	Collaborators
Store entries for one lottery	Lottery
Add and remove entrants	WaitingListEntry
Prevent duplicate entries	Entrant
Return entrants by status	
Report counts for owner monitoring	

WaitingListEntry

Responsibilities	Collaborators
Know the entrant and lottery	Entrant
Know the time the entrant joined	Lottery
Know status(waiting, selected, accepted declined, not selected)	
Change status on accept/decline	

Payment

Responsibilities	Collaborators
Know the amount and status of payment	Reservation
Process a refund	

LotteryDrawer

Responsibilities	Collaborators
Randomly select N entrants from the waiting list	WaitingList
Mark selected and not selected entries	WaitingList Entry
Draw replacements when someone declines	Messages

Lottery

Responsibilities	Collaborators
Know its campsite and the date block it covers	Campsite
Know registration open and close dates	
Know number of entrants to select	Waiting List
Know status(if it is open or not)	Lottery drawer
Accept or reject new entrants based on status	MessageService
Start the draw when requested	

GeoLocation

Responsibilities	Collaborators
Know latitude, longitude, and address	Campsite
Compute distance to another location	MapView

Entrant

Responsibilities	Collaborators
Know profile, name, email, phone #, device ID (01.09.01)	
Reserve/cancel lotteries for active campsite	
if Entrant won lottery, reserve available campsite dates	Lottery
If the user wins multiple lotteries, choose which campsite to reserve.	waitingList
If user won lottery, choose to decline reservation	
Track active reservation(s)	
Track past reservation(s)	

Give rating	
Pay/get refunded for campsite	Payment